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Sheaf representation of monoidal categories, revisited

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Every small monoidal category with universal finite joins of central idempotents is monoidally equivalent to the category of global sections of a sheaf of local monoidal categories (joint work with Chris Heunen [1]). The base space is extracted from the central idempotents of the monoidal structure itself, much like the Pierce spectrum of a ring [2]. The theorem extends the sheaf representation of toposes due to Lambek and Moerdijk [3] (see also Awodey [4]) and Stone's representation of Boolean algebras [5] from the cartesian to the monoidal setting. For Hilbert modules, it relates to Takahashi's representation by continuous fields of Hilbert spaces [6], whose fibres are quotients of our stalks, much as the residue field (fibre) is obtained from a local ring (stalk).

I will present this result and sketch more recent developments, in particular a version for closed categories [7]. There, one can work with the idempotent monads $[U, -]$ instead of the central idempotents U themselves. Meets and joins of these monads always exist, given by composites and pullbacks, so one can do away with the universal joins hypothesis. Locality of the stalks is again a disjunction property, now for a Russell–Prawitz-like disjunction defined from implication [8, 9]. Implicative semilattices [10] give an example beyond the scope of the original theorem. Finally, I will briefly outline possible connections to substructural logics, where central idempotents behave like formulas admitting weakening and an invertible contraction, such as the pure assertions of intuitionistic separation logic [11, 12], and to programming languages, where monads of this kind model security levels in information-flow type systems [13].

Keywords: Monoidal categories; closed categories; sheaf representation; central idempotents
Subject area: Monoidal categories, sheaves, and categorical logic
MSC2020: 18M05, 18F20, 18D15, 18F05, 18F70

References

[1] R. S. Barbosa and C. Heunen, Sheaf representation of monoidal categories, *Advances in Mathematics* **416** (2023), 108900.

[2] R. S. Pierce, *Modules over Commutative Regular Rings*, Memoirs of the American Mathematical Society **70**, 1967.

[3] J. Lambek and I. Moerdijk, Two sheaf representations of elementary toposes, in *The L. E. J. Brouwer Centenary Symposium*, Studies in Logic and the Foundations of Mathematics **110**, North-Holland, 1982, 275–295.

[4] S. Awodey, Sheaf representations and duality in logic, in *Joachim Lambek: The Interplay of Mathematics, Logic, and Linguistics*, Outstanding Contributions to Logic **20**, Springer, 2021, 39–57.

[5] M. H. Stone, The theory of representations for Boolean algebras, *Transactions of the American Mathematical Society* **40** (1936), 37–111.

[6] A. Takahashi, Hilbert modules and their representation, *Revista Colombiana de Matemáticas* **13** (1979), 1–38.

[7] S. Eilenberg and G. M. Kelly, Closed categories, in *Proceedings of the Conference on Categorical Algebra (La Jolla, 1965)*, Springer, 1966, 421–562.

[8] B. Russell, *The Principles of Mathematics*, Cambridge University Press, 1903.

[9] D. Prawitz, *Natural Deduction: A Proof-Theoretical Study*, Almqvist & Wiksell, 1965.

[10] W. C. Nemitz, Implicative semi-lattices, *Transactions of the American Mathematical Society* **117** (1965), 128–142.

[11] J. C. Reynolds, Separation logic: a logic for shared mutable data structures, in *Proceedings of the 17th Annual IEEE Symposium on Logic in Computer Science (LICS 2002)*, IEEE, 2002, 55–74.

[12] P. W. O'Hearn and D. J. Pym, The logic of bunched implications, *Bulletin of Symbolic Logic* **5**(2) (1999), 215–244.

[13] J. Sterling and R. Harper, Sheaf semantics of termination-insensitive noninterference, in *7th International Conference on Formal Structures for Computation and Deduction (FSCD 2022)*, LIPIcs **228**, Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022, 5:1–5:19.